Eigen-Séta v2 assessment, 2026-10-17

Authors' own analysis, not an external review. Code: `eigenx` commit `94fe2d9`. Paper: `crypto:papers/eigen-seta/` revision 2.

1. Verdict

No attack below the stated costs was found. The binding attack is still mask guessing: $\approx 2^{\lambda}$ classical chains, and $2^{\lambda/2}$ Grover queries on L1 and L3, $\geq 2^{\lambda}$ queries on L5, Q3 and Q5. Risks: (1) an effective Z[f ω]-orientation of E_A, which would give subexponential quantum key recovery at **every** set; (2) a mask-free dimension-4/8 embedding; (3) the decisional assumption.

2. Findings

#	Finding	Kind
F1	Candidate f testable from the public key alone: Kani chain on $E_A \times E_A$ of $e \approx 2\log_2 f + 3$ steps vs $E \approx 4\log_2 f$ for decapsulation. Degree-search gate columns −1.9 bits; no tier changes.	proved, measured
F2	Revision 1′s NIST categories 1/3/5 for L1/L3/L5 hold only for gate count under MAXDEPTH; L1/L3 have $2^{64.5}$ / $2^{96.5}$ quantum queries. Claim withdrawn in unqualified form.	derived
F3	Hidden orientation is the largest structural risk (Kuperberg if effective); Q sets enlarge d, not f.	literature
F4	`Fp::mul2/mul3` wrong for non-friendly moduli; `tests/field.rs::f1` red on main; no parameter set affected. Fixed.	measured
F5	SECURITY.md placed the ω-eigenbasis on $E_{D[3^F]}$ (impossible: 3 inert in $\mathbb{Q}(\sqrt{-7})$). Corrected.	derived
F6	Model filter fraction $1/\ln 2^E$ mis-justified; measured ρ is 1.6–2.3 bits higher (model conservative).	derived
F7	eigenx CI red on main at fmt/clippy. Fixed.	measured

3. Lemma 1 (public-key test)

With $\psi_A = 2\tau - tf$ (trace 0, degree $7f^2$), odd u, and $2^e - 7u^2 f'^2 = c^2 + c'^2$: the kernel $\langle (c'P', -\lambda_S P'), (c'Q', -\lambda_T Q') \rangle$ is isotropic for every candidate f', since $c'^2 + \lambda_S\lambda_T \equiv c^2 + c'^2 + 7u^2 f'^2 = 2^e$. For $f' = f$ it is Kani's kernel for $\gamma = u\psi_A + c$, which splits into $E_A \times E_A$. Lean (core, v4.34.1): `pkTest_kernel_isotropic`, `pkTest_kani_degree` pass the proof gate; the bridge from norm to isogeny degree is not formalised.

4. Measurement: run-20261017T1623Z-pk-degree-test

Shared 4-vCPU Xeon @ 2.10 GHz VM, rustc 1.97.0, no pinning. Medians [ranges].

set	E	e true	$e_{\min}$	excess	wrong (acc.)	test s	decaps s
L1	1076	569	560	9.9	105 (0)	0.79 [0.53–0.91] n=7	1.28 [1.11–1.57]
L3	1594	832	823	14.7	35 (0)	2.72 [2.64–2.78] n=3	5.54 [5.49–5.75]
L5	2106	1087	1076	14.0	18 (0)	6.84 [6.66–7.03] n=3	13.34 [11.54–14.26]

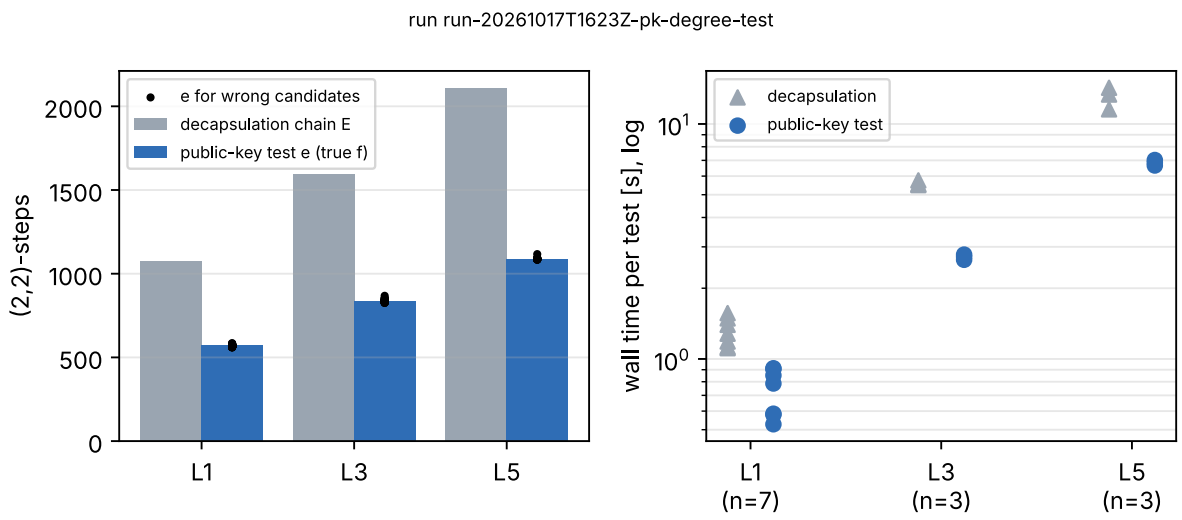


Figure 1: Chain length and wall time of the public-key test vs decapsulation; measured, run run-20261017T1623Z-pk-degree-test.

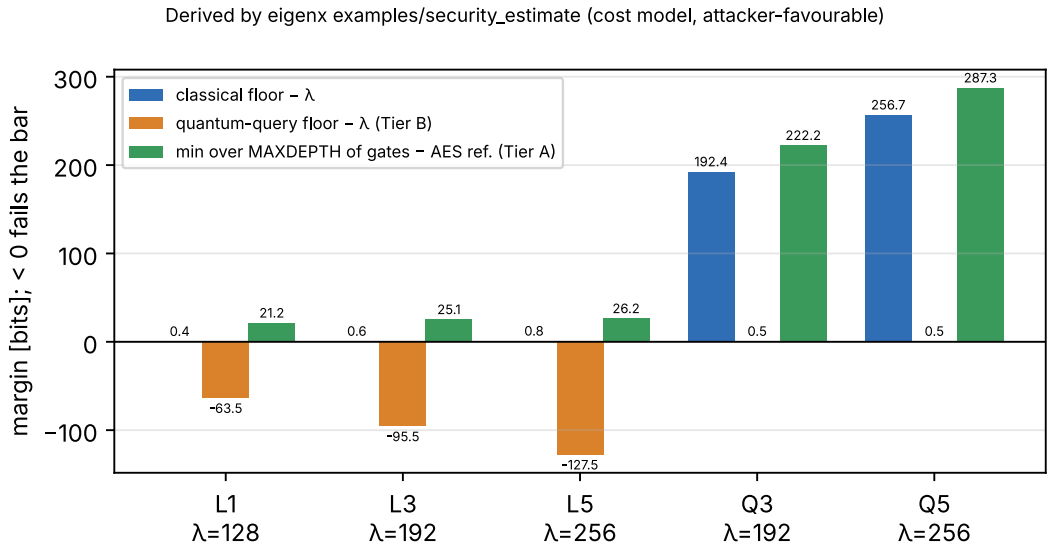


Figure 2: Margins over λ by tier, derived by the eigenx cost model (records/security_estimate.txt).

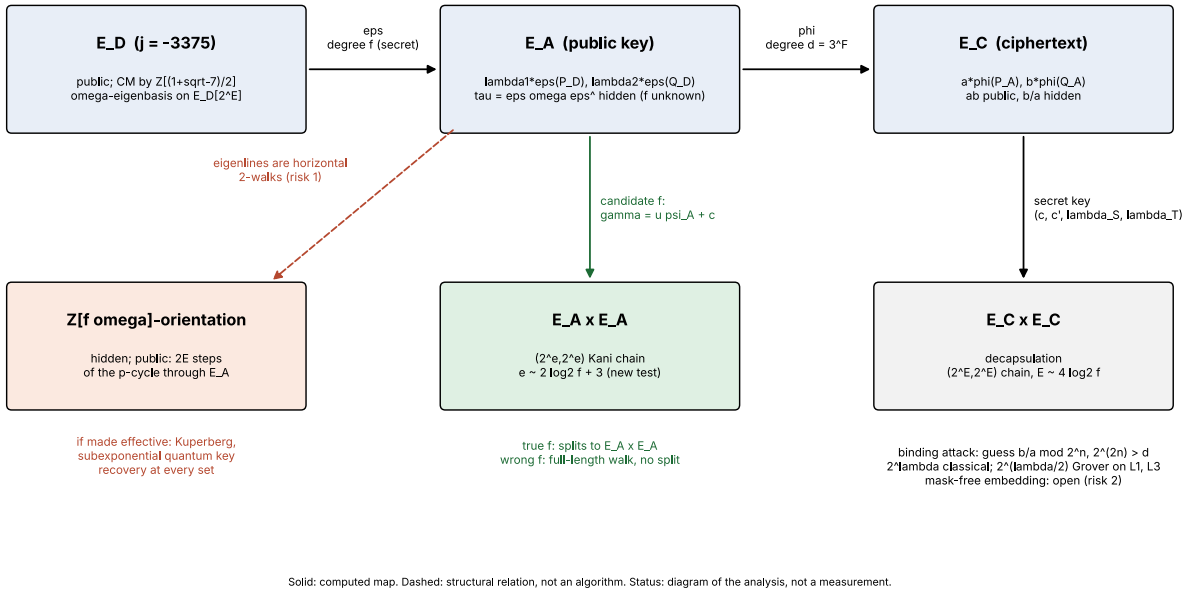


Figure 3: Structure of the scheme and the attack routes considered (diagram, not a measurement).

5. Unresolved

Risks 1–3 above (eigenx LITERATURE_DRILL L2, L6, L7, L8); external review of the 2^8 Grover-oracle constant; constant-time evidence on the theta build. Records: `records/`. Markdown twin: `REPORT.md`.